TED (21) – REVISION 2021

FIFTH SEMESTER DIPLOMA EXAMINATION IN ENGINEERING AND TECHNOLOGY

ELECTRIFICATION OF RESIDENTIAL BUILDINGS

MODEL QUESTION PAPER-SET 1

Time: 3 hours Maximum Marks: 75

PART A

I. ***Answer any 5 questions. Each question carries 3 marks***

(5 x 3 = 15 Marks).

1.	List any three types of lamp holders.	M1.03	R
2.	List any three types of wiring systems.	M1.02	R
3.	Compare LED, Fluorescent tube and incandescent lamps based on Lumen per watt.	M2.02	U
4.	Define illumination and mention its unit.	M2.01	R
5.	List three types of earthing methods.	M3.03	R
6.	Explain on what basis the number of circuits / sub circuits is determined in house wiring installations?	M3.02	U
7.	Outline any three factors to be considered while preparing an estimate.	M4.01	U

PART B

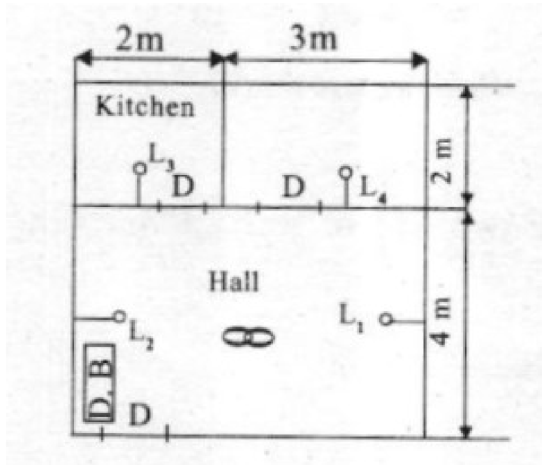
Answer ONE question from each set. Each question carries 15 marks.

(4 x 15 = 60 Marks).

II	(a).	Explain any eight general rules of electrical wiring.	M 1.02	U
	(b)	List any seven wiring accessories used for a domestic installation.	M1.03	R
		OR		
III	(a)	Explain classification of conduit wiring system and mention its merits and demerits.	M1.02	U
	(b)	Recall different types of cables used in internal wiring.	M1.04	R

IV		A room of 12m X 8m is to be illuminated with an average illumination of 80 lux on the working plane. Utilization factor is 0.5 and depreciation factor is 0.8. Lamp efficiency is 40 lumen/watt. Find the number of lamps required to illuminate the room. Also draw a schematic diagram showing the disposition of lamps.	M2.04	A
		OR		
V	(a).	Explain different types of lighting schemes.	M2.03	U
	(b)	Find the utilization factor of a room having dimensions 10 m x 6 m. It is to be illuminated by 9 lamps with a uniform illumination of 100 lux. Take 1500 lumens as the output of each lamp.	M2.01	A
VI	.	Estimate the list of materials and cost required for the pipe earthing scheme with a neat sketch.	M3.04	A
		OR		
VII	(a)	Explain circuits and sub-circuits.	M3.02	U
	(b)	Determine the number of sub-circuits for the following loads : lamps 40 W 8 Nos., fan 60 W 5 Nos., 5 A socket 60 W 8 Nos., Refrigerator 500 W 1 No., Electric heater 1000 W 1 No.	M3.02	A

VIII

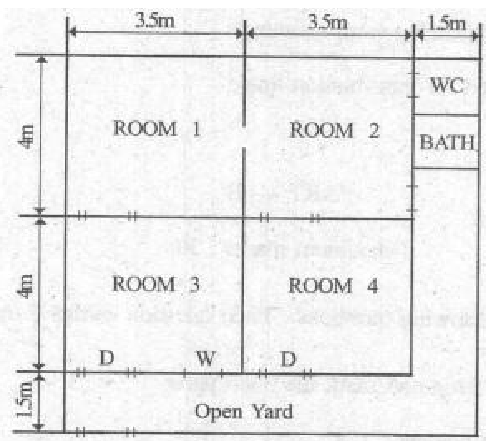


Estimate the quantity of material and its cost for the surface conduit system of wiring in a house as per the given plan. Provide one socket in the kitchen and hall. Wall thickness is 30 cm and ceiling height is 3.5 m. Assume missing data if any.

M4.04 | A

OR

IX



Location	*Light (60W)*	*Fan (80W)*	*5A Socket (80W)*	*15A Socket (1000W)*
Room-1	1	1	1	1
Room-2	1	1	1	1
Room-3	2	1	1	-
Room-4	2	1	1	-
Open Yard	1	-	-	-
WC & bath	1	-	-	-

Estimate the material and cost of a residential building using surface conduit system as per the plan.

M4.04 | A

BLUE PRINT

Mark Distribution

Module	Hr / Module	(hi / ΣHi) * 141	TYPE OF QUESTIONS					
			PART A		PART B		TOTAL	
			No of Questions	Marks	No of Questions	Marks	No of Questions	Marks
I	14	34.03	2	6	2	30	**4**	**36**
II	15	36.47	2	6	2	30	**4**	**36**
III	15	36.47	2	6	2	30	**4**	**36**
IV	14	34.03	1	3	2	30	**3**	**33**
Total	**58**	**141**	**7**	**21**	**8**	**120**	**15**	**141**

Cognitive Level Wise Question Analysis

Mark Distribution

Cognitive Level	% Marks	Marks	TYPE OF QUESTIONS					
			PART A		PART B		TOTAL	
			No of Questions	Marks	No of Questions	Marks	No of Questions	Marks
R	30	42.3	4	12	1	15	**5**	**27**
U	50	70.5	3	9	2	30	**5**	**39**
A	20	28.2	0	0	5	75	**5**	**75**
Total	**100**	**141**	7	**21**	8	**120**	**15**	**141**

<u>Question Wise Analysis</u>

Q.No	Module Outcome	Cognitive Level	Marks	Time
I.1	M1.03	R	3	7
I.2	M1.02	R	3	7
I.3	M2.02	U	3	7
I.4	M2.01	R	3	7
I.5	M3.03	R	3	7
I.6	M3.02	U	3	7
I.7	M4.01	U	3	7
II.a	M 1.02	U	8	19
II.b	M1.03	R	7	17
III.a	M1.02	U	8	19
III.b	M1.04	R	7	17
IV.	M2.04	A	15	36
V.a	M2.03	U	8	19
V.b	M2.01	A	7	17
VI	M3.04	A	15	36
VII.a	M3.02	U	5	12
VII.b	M3.02	A	10	23
VIII	M4.04	A	15	37
IX	M4.04	A	15	37
Total			141	338

FIFTH SEMESTER DIPLOMA EXAMINATION IN ENGINEERING AND TECHNOLOGY

ELECTRIFICATION OF RESIDENTIAL BUILDINGS

MODEL QUESTION PAPER-SET 2

Time: 3 hours Maximum Marks: 75

PART A

I *Answer any 5 questions. Each question carries 3 marks* ***(5 x 3 = 15 Marks)***

1.	List any three electrical safety precautions.	M1.01	U
2.	Write the functions of a) RCCB b) MCB c) MDB	M1.03	R
3.	Define luminous flux and mention its unit	M2.01	R
4.	Explain laws of illumination.	M2.04	U
5.	Draw any 3 standard electrical symbols and mention it.	M3.01	R
6.	Explain the purpose of earthing.	M3.03	U
7.	Explain the needs of star labelling in appliances.	M4.03	U

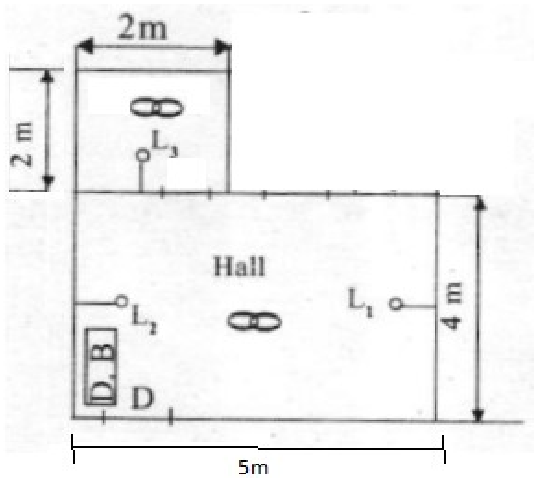
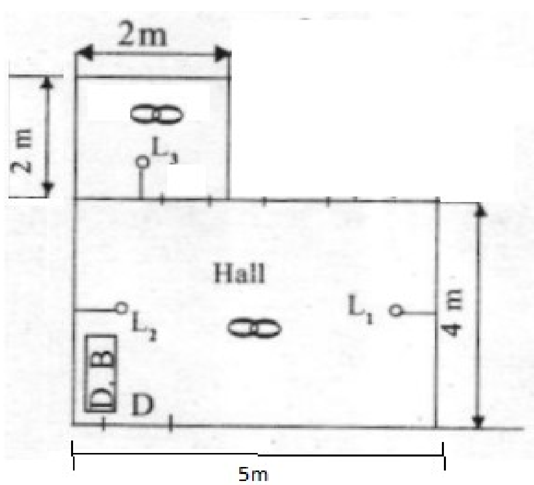
PART B

Answer ONE question from each set. Each question carries 15 marks.

(4 x 15=60 marks)

II	(a)	Draw and explain the internal distribution board for residential buildings.	M1.03	U
	(b)	Write the procedure to be taken when an accident due to electric shock occurs.	M1.01	U
		OR		
III	(a)	Explain the criterion for selection of cable.	M1.04	U
	(b)	Write any eight general rules of electrical wiring	M1.02	U
IV		A room of 20m X 10m is to be illuminated by 40 watts fluorescent lamps of lumen output of 2700 lumens.The average illumination required at the room is 300 lux. Utilization factor is 0.6 and depreciation factor is 1.25. Find the number of lamps required to illuminate the room. Also draw a schematic diagram showing the disposition of lamps.	M2.04	A
		OR		
V		An incandescent lamp hangs from the ceiling of a room. The illuminance received on a small horizontal screen lying on a bench 2m vertically below the lamp is 60 lux. Calculate the illuminance at a point when the screen is moved horizontally a distance of 1.5m along the bench	M2.04	A

VI		Explain plate earthing scheme with a neat sketch and estimate the materials required for plate earthing.	M3.04	U
		OR		
VII	(a)	Explain the design of light and power sub-circuits	M3.02	U
	(b)	A building has a total light load of 3000 watts consisting of 40 light points and power load of 5000 watts consisting of 5 points. Determine the number of sub circuits needed in the installation.	M3.02	A

VIII		Estimate the quantity of material and its cost for the surface conduit system of wiring in a house as per the given plan. Provide one socket in the room and hall. Ceiling height is 3.5 m. Assume missing data if any.Light 40W, fan 60W, socket 100W 	M4.04	A
		OR		
IX		Estimate the material and cost of a residential building using a surface conduit system as per the plan. 	M4.04	A

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Mark Distribution

Module	**Hr / Module**	**(hi / ΣHi) * 141**	**TYPE OF QUESTIONS**					
			PART A		**PART B**		**TOTAL**	
			No of Questions	Marks	No of Questions	Marks	No of Questions	Marks
I	14	34.03	2	6	4	30	**6**	**36**
II	15	36.47	2	6	2	30	**4**	**36**
III	15	36.47	2	6	3	30	**5**	**36**
IV	14	34.03	1	3	2	30	**3**	**33**
Total	**58**	**141**	**7**	**21**	**11**	**120**	**18**	**141**

Cognitive Level Wise Question Analysis:

Mark Distribution

Cognitive Level	**% Marks**	**Marks**	**TYPE OF QUESTIONS**					
			PART A		**PART B**		**TOTAL**	
			No of Questions	Marks	No of Questions	Marks	No of Questions	Marks
R	30	42.3	3	9	0	0	**3**	**9**
U	50	70.5	4	12	6	50	**10**	**62**
A	20	28.2	0	0	5	70	**5**	**70**
Total	**100**	**141**	7	**21**	11	**120**	**18**	**141**

Question Wise Analysis

Q.No	Module Outcome	Cognitive Level	Marks	Time
I.1	M1.01	U	3	7
I.2	M1.03	R	3	7
I.3	M2.01	R	3	7
I.4	M2.04	U	3	7
I.5	M3.01	R	3	7
I.6	M3.03	U	3	7
I.7	M4.03	U	3	7
II.a	M1.03	U	10	24
II.b	M1.01	U	5	12
III.a	M1.04	U	7	18
III.b	M1.02	U	8	19
IV	M2.04	A	15	36
V	M2.04	A	15	36
VI	M3.04	U	15	36
VII.a	M3.02	U	5	12
VII.b	M3.02	A	10	24
VIII	M4.04	A	15	36
IX	M4.04	A	15	36
Total			141	338

SL No. 3   OE 3 / Sem 6 / I

Scoring Indicators

Model Question Paper I

ELECTRIFICATION OF RESIDENTIAL BUILDINGS

Q No	**Scoring Indicators**	Split score	Sub Total	Total Score
	PART A			
I. 1	Batten, bracket, Pendent, Water-tight, Angle, Goliath Edison Screw lamp holder, Swivel, Starter holder, Fluorescent tube holder. (Any three).	**3x1**	**3**	
I. 2	Conduit Wiring, Cleat Wiring, Casing and Capping Wiring, Batten Wiring, Lead Sheathed Wiring. (Any three).	**3x1**	**3**	
I. 3	Lumen per watt of LED>CFL>Incandescent lamp.		**3**	
I. 4	It is the luminous flux received by the surface per unit area. It is represented by E. The unit of illumination is lux. Illumination E = flux / Area		**3**	
I. 5	Pipe earthing, Wire earthing, plate earthing.	**3x1**	**3**	
I. 6	The number of circuits and sub-circuits is decided on the basis of total number of points to be wired and the total load to be connected to the supply system.		**3**	
I. 7	Project location, item cost , tax rate on each item, contingency expenses, labour cost , validity period . (Any three).	**3x1**	**3**	**15**

	PART B			
II. a)	·The current rating of the cable should be slightly greater than the load (at least 1.5 times). ·Every live wire should be protected by a fuse of suitable rating. ·Every sub-circuit should be connected with the fuse distribution board. ·All metal coverings used for the protection of earth must be connected to earth. ·No switch of fuse is used in earth or neutral conductor. ·All the switches and starters should be accessible to the operator. ·Every apparatus should be provided with a separate switch. ·A caution notice should be fixed on every equipment. ·In any building light wiring and power wiring should be kept separately. ·In 3 phase 4 wire installation the load should be distributed almost equally. ·In three phase, 3 phases should be indicated by Red, Blue and Green. ·Neutral should be indicated by Black wire.	**(any eight)** **8x1**	**8**	
II. b)	Switches, lamp holders, ceiling roses, socket outlets, plug pins PVC conduits, Switch boards, PVC beds and L bows, Nut and Screws, Round block etc (any seven).	**7x1**	**7**	**15**

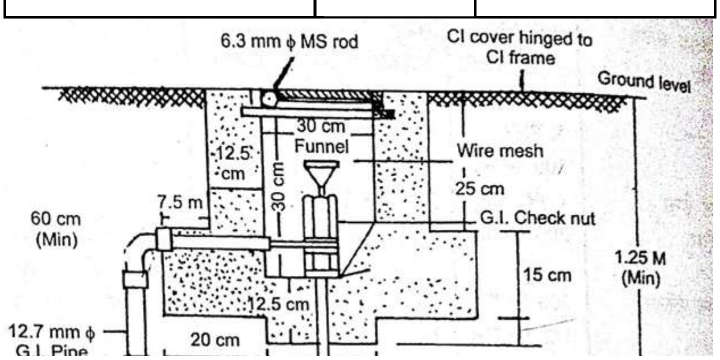
III. a)	Conduit Wiring System : It consists of VIR or PVC cables taken through rigid steel pipes known as conduits Conduit wiring system can be classified into two: a)Concealed or Recessed conduit Wiring : The conduits are embedded along walls or ceiling in plaster at the time of construction. The conduits are fixed by means of saddles or staples not more than 60mm apart. Now-a-days PVC conduits are widely being used as it is resistant to acids, alkalis, oils and moisture. Concealed conduit wiring is used in residential, commercial and public buildings. b) Surface Conduit Wiring: All conduits are fixed to the surface of the wall with the help of saddles with a fixed interval of span not more than 0.60m. PVC conduits are widely used because of its weather resistance and its life span. Surface conduit is used for factory or workshop lighting and for motor wiring. <u>Advantages</u> I.This system is waterproof 2. Replacement of effective wiring is easy 3. It is shock proof 4. It requires less time <u>Disadvantages</u> I. PVC conduits doesn't provide protection against fire hazards 2. Metal conduit wiring is very costly 3. Metal conduit wiring needs skilled labour.		**8**	

III. b)	The wires employed for internal wiring of buildings may be divided into different groups according to (i) conductor used (ii) number of cores used, (iii) voltage grading and (iv) type of insulation used. According to the conductor material used in cables, these may be divided into two classes known as copper conductor cables and aluminium conductor cables. According to the number of cores, the cables may be divided into classes known as single core cables; twin core cables; three core cables; two core with ECC cables etc. According to voltage grading the cables may be divided into two classes : (i) 250/440 volt cables and (ii) 650/1,100 volt cables. According to type of insulation the cables are of the following types : 1. Vulcanized Indian Rubber (VIR) insulated cables. 2. Tough rubber sheathed (TRS) or cab tyre sheathed (CTS) cables. 3. Lead sheathed cables. 4. Polyvinyl chloride (PVC) cables. 5. Weatherproof cables. 6. Flexible cords and cables. 7. XLPE cables.		**7**	**15**

IV	Total floor area = 12 x 8 = 96 m^2. Total flux = 80 x 96 = 7680 lumen. Total flux required on working plane = Total flux / (cu x mf) = 7680/(0.5 x 0.8) = 19200 lumen. Total flux = flux / efficiency = 19200 /40 = 480 W Number of lamps required = 480 / 40 = 12 lamps. (9 marks) Schematic diagram. (6 marks)		**15**	**15**
V. a)	<u>Direct fitting:</u> - 90-100% directed towards the working plane, 10% goes to the other direction. The height of the lamp is two thirds of the lamp spacing. Reflectors can be used. This type may produce hard shadows. <u>Semi-direct fitting:</u> - 60-90% on the working plane, 10-40% goes the other direction. Translucent reflectors can be used. This type also produces shadows. <u>General fitting:</u> - 40-60% on the working plane, 60-40% goes the other direction. Translucent reflectors of different thickness can be used. This fitting produces almost uniform light. <u>Semi-indirect fitting:</u> - 10-40% light on the working plane, remaining light goes to the upper hemisphere. Light on the working plane by the reflectivity of the ceiling and walls. This produces faint shadows. <u>Indirect fitting:</u> - 10% of the light on the working plane due the reflectivity of the walls and ceiling. This will not produce any shadows or any glare. These are used for clubs and restaurants.	**8**	**8**	
V. b)	Total lumens by lamp = 13500 lumen, Total lumen received on the working plane = area X average lumen = (10x 6) X 100 = 6000 lumen. Utilization factor =Lumens received on the working plane / lumens emitted by the source. 6000 /13500 = 0.444.	**7**	**7**	**15**

VI		**7+8**	**15**	**15**

Material	Quantity	Cost
38mmGI pipe (perforated)	2.5	500
Gl wire (8SWG)	10m	100
12.7mm GI pipe	2m	150
19 mm Gl pipe	1m	250
GI(38x 19mm) reducing socket	1	100
GI nut,bolt, washers, checknut	2 set	150
GI bend 12.7mm	2	100
GI lugs	1 box	100
Cast iron frame with hinges	1	500
Cast iron cover 30cmX30 cm	1	1000
Funnel with wire mesh	1	150
Charcoal or cock	1 sack	500
salt	1 sack	250
Cement	1 sack	400
Caution plate	1	100
Labour Charge		5000
Total		9350
10 % contingency		935
Grand Total		10285



VII. a	A Circuit is a conducting path through which either an electric current flows or is intended to flow. A sub-circuit is a path in which the amount of current or devices connected is limited to a certain value.	**2*2.5**	**5**	
VII. b	Total watts of lamp = 8 X 60 = 480 W Total watts Of fan 5 X 60 = 300 W Total watts of 5 A sockets = 8 X 60 = 480 W Refrigerator = 500 W Total load = 1760 W The number of light , fan, socket limited to ten or 800 W Therefore, no of Sub circuits = 1760 /800 = 2.2, i.e. 3. Provide one sub circuit for heater Total sub circuit = 3+1 = 4	**10**	**10**	**15**

<table>
<tr>
<td>VIII</td>
<td>Number of sub circuits :
Total watts = 520 W, so we need only one sub circuit .
Size of the cable :
Current through sub circuit = power / voltage =520 / 230 =2.26.
Size of the cable required is 1 mm^2 copper.
Length of conduit = 20 m
Cable length = Length of conduit X 3 = 60m
Labour cost = Rs. 400 per point = 400x9 = 3600 rupees.
<table>
<tr><th>Material</th><th>Quantity</th><th>Cost</th></tr>
<tr><td>DP main switch 15A, 250 V</td><td>1</td><td>1000</td></tr>
<tr><td>IC cutout 15 A , 250 V</td><td>1</td><td>250</td></tr>
<tr><td>Flush type fuse unit 5A,250V</td><td>1</td><td>250</td></tr>
<tr><td>PVC conduit 18mm</td><td>20m</td><td>300</td></tr>
<tr><td>PVC conduit 12mm</td><td>1m</td><td>10</td></tr>
<tr><td>1mm2 cable</td><td>63m</td><td>750</td></tr>
<tr><td>Switches 5 A , 250 V</td><td>7</td><td>420</td></tr>
<tr><td>2-pin sockets 5 A , 250 V</td><td>2</td><td>120</td></tr>
<tr><td>Ceiling rose</td><td>4</td><td>40</td></tr>
<tr><td>Lamp holders</td><td>4</td><td>80</td></tr>
<tr><td>Switch boards</td><td>5</td><td>600</td></tr>
<tr><td>Wooden gutties</td><td>3 box</td><td>120</td></tr>
<tr><td>Saddles</td><td>1 box</td><td>40</td></tr>
<tr><td>Nails</td><td>2 kg</td><td>100</td></tr>
<tr><td>Cement</td><td>5 kg</td><td>100</td></tr>
<tr><td>Earthwire 16SWG copper</td><td>1 kg</td><td>300</td></tr>
<tr><td>Earth set</td><td>1 set</td><td>5000</td></tr>
<tr><td>Labour cost</td><td></td><td>3600</td></tr>
<tr><td>Total</td><td></td><td>13080</td></tr>
<tr><td>10% contingency</td><td></td><td>1308</td></tr>
<tr><td>Grand Total</td><td></td><td>14388</td></tr>
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<tr>
<td>IX</td>
<td>Total light wattage = 1120 watts.
No of light sub circuits =2,
Total power wattage = 2000 watts
No of power sub circuits = 2
1 mm^2 cable can be used for wiring.
Length of conduit for sub circuit 1 =19m.
Length of conduit for sub circuit 2 = 22 m.
Length of cable required for sub circuit = (19+22)x 3= 123 m
<table>
<tr><th>Material</th><th>Quantity</th><th>Cost</th></tr>
<tr><td>IC cutout 15 A , 250 V</td><td>1</td><td>300</td></tr>
<tr><td>DP main switch 15A, 250 V</td><td>1</td><td>1000</td></tr>
<tr><td>Flush type fuse unit 5 A , 250 V</td><td>1</td><td>250</td></tr>
<tr><td>Flush type fuse unit 15 A , 250 V</td><td>1</td><td>500</td></tr>
<tr><td>PVC conduit 25mm</td><td>60 m</td><td>400</td></tr>
<tr><td>1mm2 Cable</td><td>123 m</td><td>800</td></tr>
<tr><td>Switches 5 A , 250 V</td><td>12</td><td>400</td></tr>
<tr><td>3-pin sockets 5 A , 250 V</td><td>4</td><td>200</td></tr>
<tr><td>Ceiling rose</td><td>6</td><td>120</td></tr>
<tr><td>Lamp holders</td><td>6</td><td>150</td></tr>
<tr><td>Switch boards</td><td>8</td><td>800</td></tr>
<tr><td>Saddles 25mm</td><td>4 box</td><td>400</td></tr>
<tr><td>Wooden gutties</td><td>12</td><td>200</td></tr>
<tr><td>Nails</td><td>2.5 Kg</td><td>200</td></tr>
<tr><td>Cement</td><td>5 Kg</td><td>100</td></tr>
<tr><td>Earth wire 16SWG copper</td><td>1 Kg</td><td>300</td></tr>
<tr><td>Earth set</td><td>1</td><td>5000</td></tr>
<tr><td>Labour cost</td><td></td><td>10000</td></tr>
<tr><td>Total</td><td></td><td>21040</td></tr>
<tr><td>10% contingency</td><td></td><td>2104</td></tr>
<tr><td>Total</td><td></td><td>23144</td></tr>
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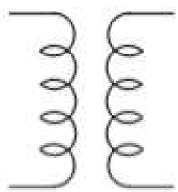
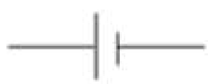
SL No. 3   OE 3 / Sem 6 / II

Scoring Indicators

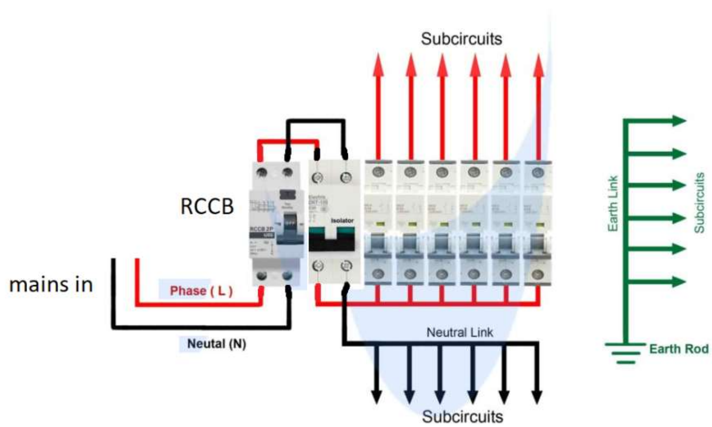
Model Question Paper II

ELECTRIFICATION OF RESIDENTIAL BUILDINGS

Q No	Scoring Indicators	Split score	Sub Total	Total score
	PART A			
1	• Avoid water at all times when working with electricity. • Never use equipment with frayed cords, damaged insulation or broken plugs. • If you are working on any receptacle at your home then always turn off the mains. • Always use insulated tools while working.	3x1	3	
2	RCCB: (Residual Current Circuit Breaker)It disconnects the circuit whenever there is leakage of current flow through the Human body or the current is not balanced between the phase conductor. MCB: (Miniature Circuit Breaker) It automatically switches OFF electrical circuit during any abnormal condition in the electrical network such as overload & short circuit conditions. MDB : Divides an electrical power feed into subsidiary circuits and thus provides independent protective devices to these sub circuits.	3x1	3	
3	Luminous Flux is luminous energy per unit time (dQ/dt) that is radiated from a source over visible wavelengths. More specifically, it is energy radiated over wavelengths sensitive to the human eye, from about 330 nm to 780 nm. SI unit is lumen.	2 1	3	

4	Inverse Square Law: Intensity of Illumination produced by a point source varies inversely as square of the distance from the source. $E \alpha I / d^2$ Lambert cosine law: The illumination at any point on a surface is proportional to the cosine of the angle between the normal at that point and the direction of light flux $E \alpha \; I \cos \theta / d^2$ (with necessary figures)	1.5 1.5	**3**	
5	 Transformer  Motor  Single Cell  Fuse	3 x 1	**3**	
6	● ·To protect us from an electric shock ● ·To protect buildings, machinery & appliances under fault conditions. ● ·Helps the protective device (either a circuit-breaker or fuse) to switch off the electric current to the circuit that has the fault. ● ·To provide stable platform for operation of sensitive electronic equipment ● ·To ensure that all exposed conductive parts do not reach a dangerous potential.	3 x 1	**3**	

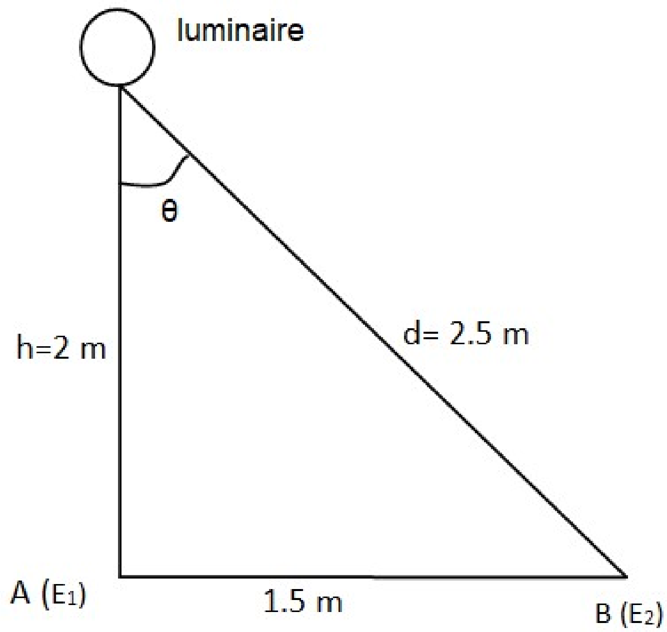
7	● Star labelling is meanly required to recognize quality of product ● Star labelling is also required to determine life and efficiency of the product. ● Star labels identify the percentage of energy conservation products.	3 x 1	**3**	**15**

	PART B			
II. 1	 (Explanation)	**6** **4**	**10**	
II. 2	● Separate the Person From Current's Source by ensuring your safety ● Call ambulance ● Do CPR, if Necessary ● Check for Other Injuries and provide first aid ● Doctor check up the person for any burns, fractures, dislocations, and other injuries	**5 x 1**	**5**	**15**

III.1	·Voltage grade · Load factor (current carrying capacity) - Short circuit withstand capacity - Voltage drop - Temperature rise - Insulation to withstand abnormal voltage as per standards - Power losses vs cost of installation.	7x1	**7**	
III. 2	- The current rating of the cable should be slightly greater than the load (at least 1.5 times). - Every live wire should be protected by a fuse of suitable rating. - Every sub-circuit should be connected with the fuse distribution board. - All metal coverings used for the protection of earth must be connected to earth. - No switch of fuse is used in earth or neutral conductor. - All the switches and starters should be accessible to the operator. - Every apparatus should be provided with a separate switch. - A caution notice should be fixed on every equipment. - In any building light wiring and power wiring should be kept separately. - In 3 phase 4 wire installation the load should be distributed almost equally. - In three phase, 3 phases should be indicated by Red, Blue and Green. - Neutral should be indicated by Black wire.	8x1	**8**	**15**

IV	Total floor area = 20 x 10 = 200 m^2. Required illumination = 300 lux Lumen output of the lamp (O) = 2700 lumens Utilization factor (cu) = 0.6 Depreciation factor = 1.25 mf = 1/ (Depreciation factor) = 0.8 Number of lamps = Total flux / O x (cu x mf) = 300 x 200/(0.6 x 0.8)x 2700 = 46 (9 marks) Schematic diagram. (6 marks)	15	**15**	**15**

V



Illuminance at point A = E_1 = 60 lux
Distance between the screen and the lamp = h = 2m
When the screen is moved horizontally 1.5 m distance between screen and lamp

$d=\sqrt{2^2 + 1.5^2} = 2.5\text{m}$

Illuminance at point A = $E_1 = \frac{I}{h^2}$

$I = E_1 \times h^2$
$= 60 \times 2^2$
$= 240$ lumens

$\text{Cos}\ \theta = \frac{h}{d}$

$E_2 = \frac{I}{d^2}\text{Cos}\ \theta$

$E_2 = \frac{I}{d^2}\frac{h}{d}$

$= \frac{240 \times 2}{2.5^3}$

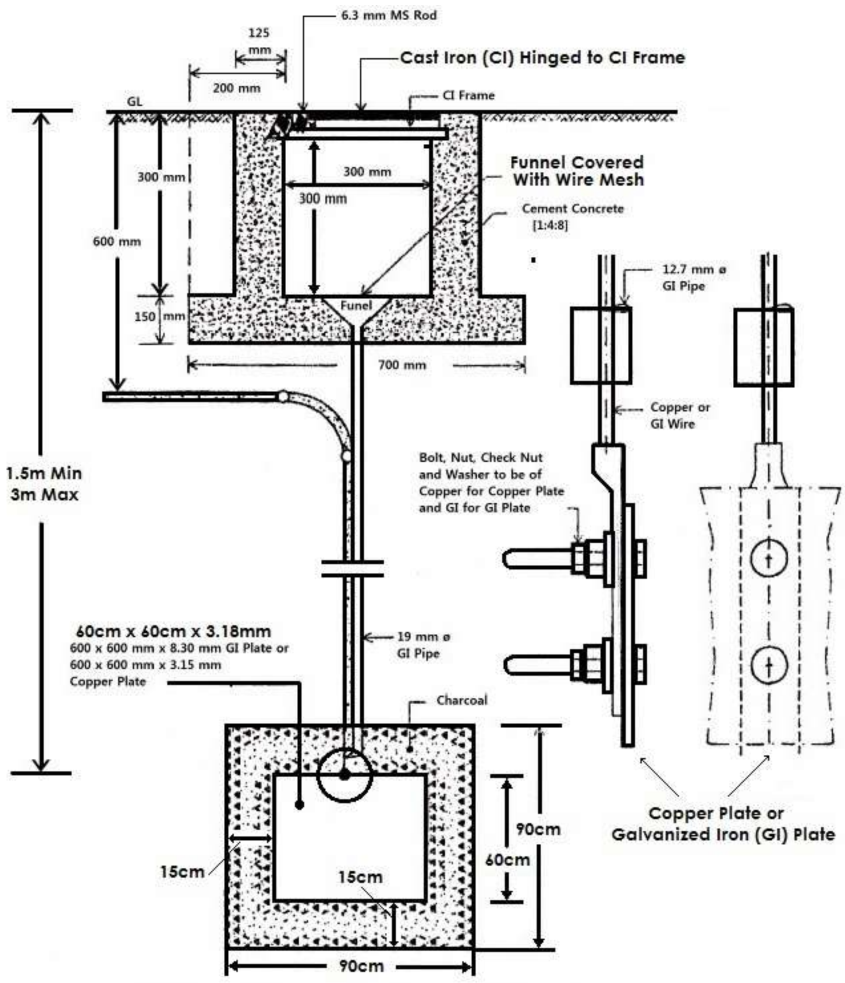
=30.72 lux

15

15

15

VI



Material	Qty	Cost
19 mmGI pipe	1m	250
600mm x 600mm x8.3mm GI plate	1	2000
GI lug	1 box	200
GI nut,bolt,washers,checknut	2 set	350
GI wire(8 SWG)	10m	200
Cast iron frame with hinges	1	650
Cast iron cover 30cm x 30cm	1	1500
funnel with wire mesh	1	200
Characol	1 sack	750
Salt	2 sack	250
Cement	3 sack	650
Caution plate	1	500
Labour charge		8000
Total		15500

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VII.a	Light sub circuit: Load on each sub circuit shall be restricted to 800 W or max 10 outlets. Power sub circuit: Load on each power sub circuit shall be restricted to 2000 W or max 2 outlets.	2*2.5	**5**	
VII.b	Total light load= 3000 W Max amount of light load that can be accommodated in one light sub circuit = 800 W Therefore number of light sub circuits = 3000/800 = 3.75 = app 4 Max Number of light points in a given sub circuit = 10 So there can be 4 x 10 = 40 light points can be connected. Power Load: Total power load = 5000W There can be max of 2000W in one sub circuit. So number of power sub circuits required = 5000/2000 =2.5=app 3 So max number of power points possible in power sub circuit is 2. Hence there can be 3 x 2 = 6 power points. Since there are only 5 power points so 3 sub circuits are required for power load.	10	**10**	**15**

VIII

Number of sub circuits :
Total watts = 440 W, so we need only one sub circuit .
Size of the cable :
Current through sub circuit = power / voltage =440 / 230 =1.91A.
Size of the cable required is 1 mm^2 copper.
Length of conduit = 15 m
Cable length = Length of conduit X 3 = 45m
Labour cost = Rs. 300 per point = 300x7 = 2100 rupees.

10

MATERIAL	QTY	COST
DP main switch 15A,250V	1	1000
IC cutout 15A, 250V	1	250
Flush type fuse unit 5A,250V	1	250
PVC conduit 18mm	20m	300
PVC conduit 12mm	1m	10
1mm sq cable	63m	750
switches 5A,250V	7	420
2-pin sockets 5A , 250V	2	120
ceiling rose	4	40
Lamp holder	4	80
switch board	5	600
saddles	1box	40
nails	2kg	100
cement	5kg	100
earthwire 16SWG copper	1kg	300
Earth set	1 set	5000
Labour cost		2100
Total		10735
10% contingency		1073.5
Grand Total		11808.5

5

15

15

<table>
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<td>IX</td>
<td>
Total light circuit wattage = 780 W

Total number of points in light circuit = 11
So 2 sub circuits are required for light circuit.
Total power circuit wattage = 2000W
So 1 sub circuit for power circuit.

1 mm^2 cable can be used for wiring.
Length of conduit for sub circuit 1 =19m.
Length of conduit for sub circuit 2 = 22 m.
Length of cable required for sub circuit = (19+22)x 3= 123 m

Length of conduit for power sub circuit = 8m
Length of cable required for power sub circuit = 30m

<table>
<tr><th>MATERIAL</th><th>QTY</th><th>COST</th></tr>
<tr><td></td><td></td><td></td></tr>
<tr><td>DP main switch 15A,250V</td><td>1</td><td>1000</td></tr>
<tr><td>IC cutout 15A, 250V</td><td>1</td><td>250</td></tr>
<tr><td>Flush type fuse unit 5A,250V</td><td>2</td><td>500</td></tr>
<tr><td>PVC conduit 18mm</td><td>41m</td><td>615</td></tr>
<tr><td>PVC conduit 12mm</td><td>9m</td><td>90</td></tr>
<tr><td>1mm sq cable</td><td>123m</td><td>3800</td></tr>
<tr><td>2.5mm sq cable</td><td>30</td><td>600</td></tr>
<tr><td>switches 5A,250V</td><td>11</td><td>660</td></tr>
<tr><td>switches 15A,250V</td><td>2</td><td>200</td></tr>
<tr><td>3-pin sockets 5A , 250V</td><td>3</td><td>180</td></tr>
<tr><td>3-pin sockets 15A , 250V</td><td>2</td><td>500</td></tr>
<tr><td>ceiling rose</td><td>8</td><td>80</td></tr>
<tr><td>Lamp holder</td><td>5</td><td>100</td></tr>
<tr><td>switch board</td><td>3</td><td>360</td></tr>
<tr><td>saddles</td><td>1box</td><td>40</td></tr>
<tr><td>nails</td><td>2kg</td><td>100</td></tr>
<tr><td>cement</td><td>5kg</td><td>100</td></tr>
<tr><td>earthwire 16SWG copper</td><td>1kg</td><td>300</td></tr>
<tr><td>Earth set</td><td>1 set</td><td>5000</td></tr>
<tr><td>Labour cost</td><td></td><td>4000</td></tr>
<tr><td>Total</td><td></td><td>18475</td></tr>
<tr><td>10% contingency</td><td></td><td>1848</td></tr>
<tr><td>Grand Total</td><td></td><td>20323</td></tr>
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